

QRL on Multi-Arm Bandit and Lunar Lander Problems

Soren Schultz, Margaret Goldstein, Salma Bhar, Shouren Wang, Xianxuan Long

Advisor: Professor Jesse Berezovsky, Instructors: Professors Jesse Berezovsky, Christos Papachristou, Stanislaw Szarek, Shuai Xu

Introduction

Quantum Reinforcement Learning (QRL) merges quantum physics and reinforcement learning to develop faster, more efficient algorithms.

This project applies **Quantum Reinforcement Learning (QRL)** to **Multi-Armed Bandit (MAB)** and **full RL tasks** (CartPole, LunarLander), exploring whether quantum methods can:

- Improve **exploration efficiency** through **quantum superposition**
- Achieve **faster convergence**
- Reduce **trainable parameters**
- Enable **stable policy learning**

We compare **classical methods** (ϵ -greedy classical RL) with **quantum VQC-based agents** to evaluate performance gains and engineering potential.

Methodology

Classical Reinforcement Learning Approach

- **Multi-Armed Bandit (MAB)** with ϵ -greedy algorithm:
 - Random exploration (ϵ probability) vs best known arm ($1-\epsilon$ probability)
 - Reward Distribution: Normal distribution centered around [1.0, 2.0, 3.0]
 - Tested ϵ values: 0.1, 0.05, 0.01
 - Evaluation Metric: Cumulative average reward over time

Quantum Reinforcement Learning Approach

- **Variational Quantum Circuits (VQC):**
 - Parameterized rotation gates + entanglement gates
 - Policy networks for CartPole (4-qubit VQC) and LunarLander (8-qubit VQC)
 - Two architectures:
 - **VQC-DQN:** Quantum policy, classical value updates
 - **Hybrid A2C:** Quantum actor + classical critic

Interdisciplinary Advancements

Engineering Perspective

The multi-armed bandit framework addresses a core engineering need: making optimal decisions under uncertainty with limited data.

- **Applications**
 - **Wireless Communication**
 - **Robotics and Control Systems**

Physics Perspective

- **Information maximization** → Physics principle in which decisions are made to maximize the information gained through each decision by aiming to minimize the entropy of the system
- **Quantum Superposition** → System can exist in multiple states allowing for simultaneous exploration which can reduce decision making time

Computer Science Perspective

- QRL brings provable quadratic time advantages, exponentially more compact policy/state representations, and a unifying **differential** programming framework for hybrid quantum-classical agents

Results

Classical Reinforcement Learning (ϵ -Greedy)

Learning Performance

- Using an ϵ -greedy algorithm, the agent gradually learned to approach the highest true reward (≈ 3.0) across 100,000 trials but the speed and stability of convergence varied with ϵ .
- A strategic balance between exploration (random selection) and exploitation (best-known arm) enabled convergence.

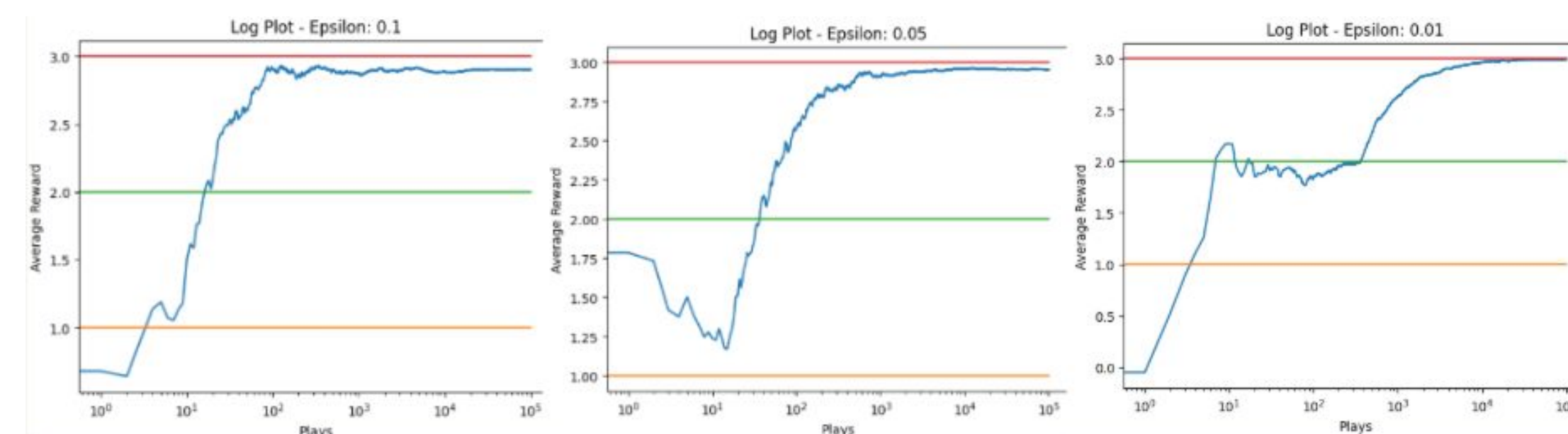


Fig. 1: Comparison of Classical ϵ -greedy Policies Learning Progress in Log Plots

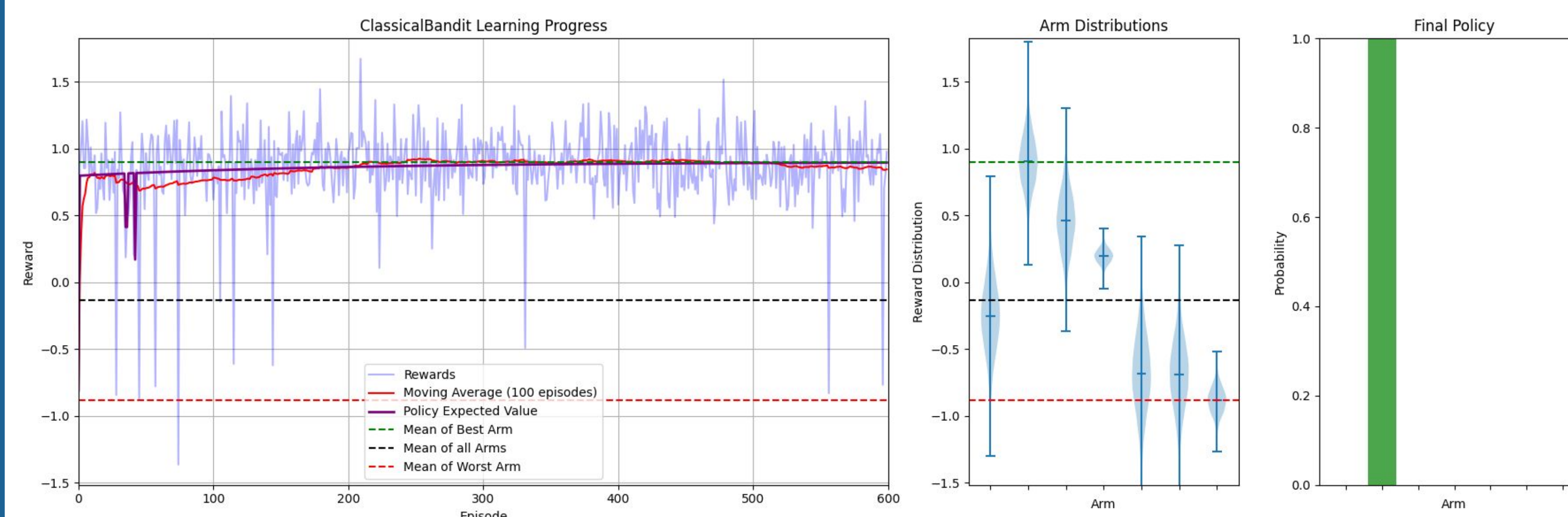


Fig. 2: Classical ϵ -greedy Learning Progress, True Arm Distributions, and Final Policy

Quantum Reinforcement Learning (REINFORCE algorithm)

Learning Performance

- Using a similar ϵ -greedy algorithm based on the REINFORCE algorithm, the agent repeatedly sampled actions from a learned distribution.
- The strain of exploration was split between the random sampling from the learned distribution and the ϵ chance of uniformly selecting an action.
- Gradient clipping was introduced to prevent significant updates to the parameters
- This induced a much smoother, balanced trend as the agent proportionally valued high-value arms that weren't necessarily the optimal choice

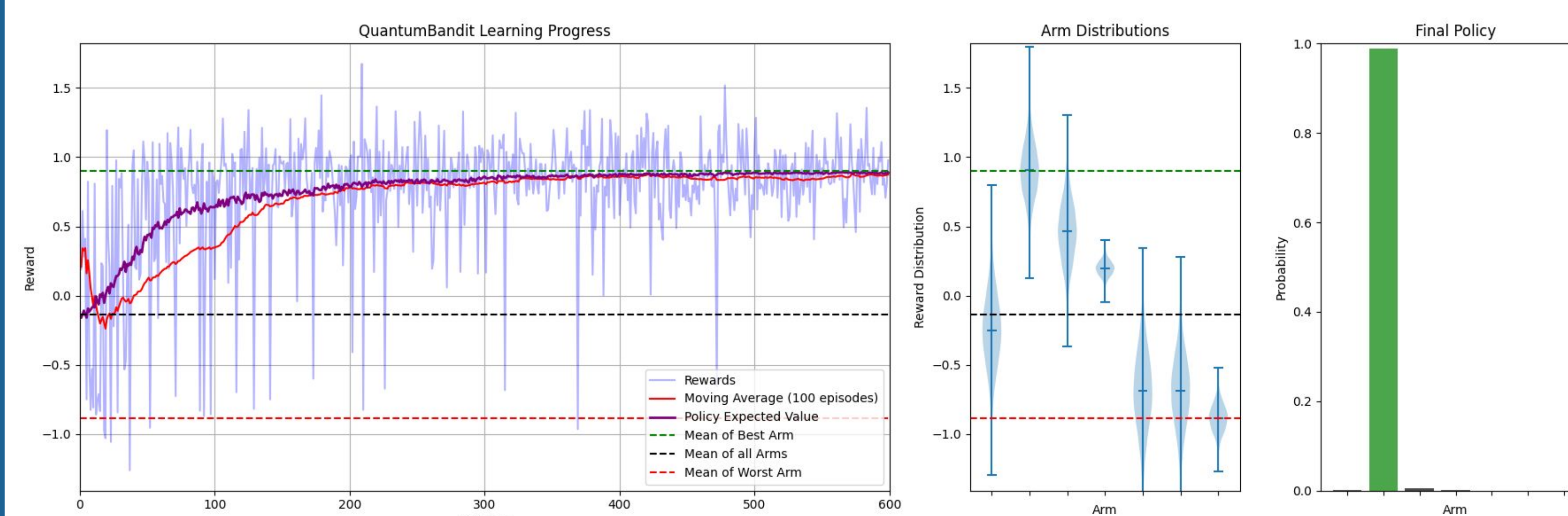


Fig. 3: Quantum Learning Progress, True Arm Distributions, and Final Policy

Extend QRL to MDP-based RL Tasks

Learning Performance

- We extended QRL from classical bandits task to markov decision process (MDP) based tasks — Cart Pole and Lunar Lander.
- QRL reduced the number of trainable parameters up to 90% compared to classical RL model.
- The training curves indicate that QRL achieved performance comparable to that of classical RL methods, and even exhibited more stable convergence.

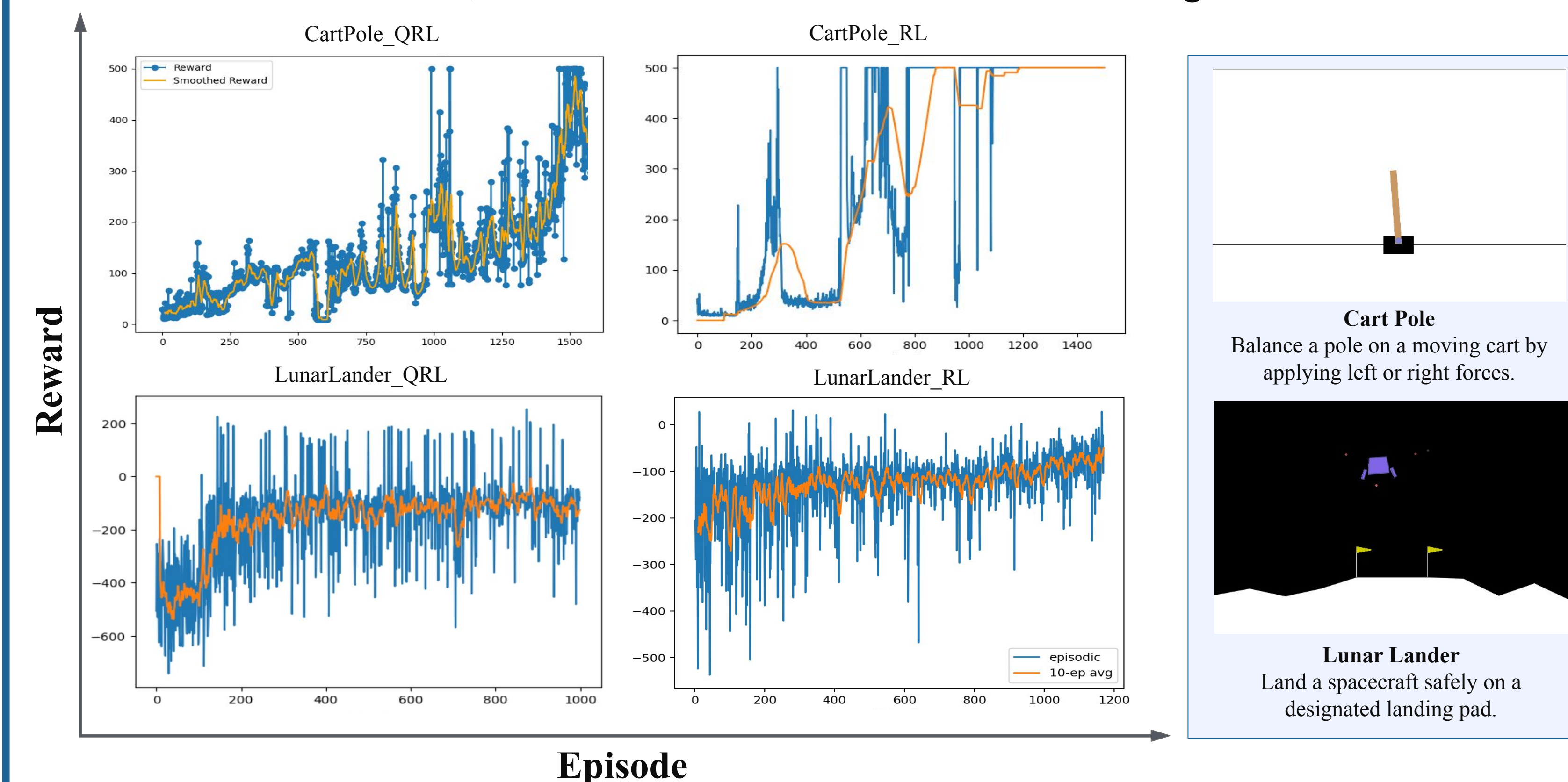


Fig. 4: QRL for MDP-based RL task — Cart Pole and Lunar Lander

Conclusion

- **Classical RL (ϵ -greedy):** Reliable convergence, with trade-offs between exploration and early reward.
- **Quantum RL Approach:** The QuantumBandit uses dramatically fewer tunable numbers— $O(\log(k))$ vs. $O(k)$ —making it parameter-efficient. However, that advantage is offset by heavier simulation cost and noisier gradient estimates. In MDP-based tasks, compared to classical RL algorithms, QRL reduces the trainable parameters up to 90%, achieves comparable performance and convergence.
- **Physics Perspective:** Quantum superposition accelerated **information gain** and **entropy reduction**.
- **Outlook:** Quantum RL, despite current hardware limits, shows strong promise for **fast, data-efficient decision-making** in engineering systems.

References

- [1] GeeksforGeeks, "Epsilon greedy algorithm in reinforcement learning," [Online]. Available: <https://www.geeksforgeeks.org/epsilon-greedy-algorithm-in-reinforcement-learning/>. [Accessed: April 2025].
- [2] GeeksforGeeks, "Multi-armed bandit problem in reinforcement learning," [Online]. Available: <https://www.geeksforgeeks.org/multi-armed-bandit-problem-in-reinforcement-learning/>. [Accessed: April 2025].
- [3] T. Bonald, "Multi-armed bandit algorithms," Télécom ParisTech, [Online]. Available: <https://perso.telecom-paristech.fr/bonald/documents/bandit.pdf>. [Accessed: April 2025].
- [4] A. Barbier-Chebbah, "Information maximization for a broad variety of multi-armed bandit games," *arXiv preprint, arXiv:2503.15962*, 2025.
- [5] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA: MIT Press, 2018.
- [6] R. Kumar, T. R. Ashwin, and P. Banerjee, "Optimization of reinforcement learning using quantum computation," 2022.
- [7] E. Farhi and H. Neven, "Classification with quantum neural networks on near term processors," 2018.
- [8] L. Li, W. Chu, J. Langford, and R. E. Schapire, "A contextual-bandit approach to personalized news article recommendation," in *Proc. 19th Int. Conf. World Wide Web (WWW)*, 2010, pp. 661–670.
- [9] H. Mao, M. Alizadeh, I. Menache, and S. Kandula, "Resource management with deep reinforcement learning," in *Proc. 15th ACM Workshop Hot Topics in Networks (HotNets '16)*, 2016, pp. 50–56.
- [10] Y. Abbasi-Yadkori and C. Szepesvári, "Regret bounds for the adaptive control of linear quadratic systems," in *Proc. 24th Annu. Conf. Learning Theory (COLT)*, 2011.
- [11] Anonymous, "Recent advances in quantum RL," *arXiv preprint*, [Online]. Available: <https://arxiv.org/pdf/2409.05846>. [Accessed: April 2025].
- [12] Anonymous, "Quantum approaches to reinforcement learning," *arXiv preprint*, [Online]. Available: <https://arxiv.org/pdf/2103.05577>. [Accessed: April 2025].